

# Project Log

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## Introduction

This document contains the log of activities and hours spent on the individual project for the course individualProject for Avans.

In this course the student needs to make a project that incorporates the graphical card to do some calculations. The student needs to use OpenGL or OpenCL to do this.

## Goal

The goal for this project is to use openCL to make a fluid simulation where water can be defined in a square space and the graphics card then calculates how the water flows. The focus first will be to make this in 2 dimensions. If this is done and there is time left the project will be expanded to 3 dimensions.

## Log of Activities

### 1. Read brightspace and init project

- **Activity** Read the brightspace course
- **Activity** Setup code environment
- **Activity** Setup git repository
- **Activity** Write template for this log

### 2. Google what is OpenCL

- **Activity** Read OpenCL landig page
- **Activity** Read What is OpenCL

### 3. Google openCL tutorial

- **Activity** Found getting started linux and followed part of tutorial with own knowledge to see if my envirement on new arch linux is working correctly.

- **Activity** Installed openCL headers
- **Activity** Followed Arch wiki to install openCL
- **Note** First program did not work. Had to google around and after rereading Arch wiki found I also had to install openc-headers package instead of only openc-nvidia.
- **Note** Added user to the video group using "sudo usermod -aG video sean", not sure if this was necessary

#### 4. Trying to use c++

- **Activity** Found OpenCL-CLHPP and looked at the example
- **Founding** Saw I have to add extra find package to my cmakeLists.txt (OpenCLHeaders, OpenCLICDLoader and OpenCLHeaderCpp)
- **Activity** After searching and trying things for about an hour and nothing working I remembered I can just add the raw hpp file to my project and use it that way.

#### 5. Search and follow basic tutorial of openCL

- **Activity** Found and followed Simple start with OpenCL and C++
- **Note** In the tutorial a device is selected. At first there was no known device. After searching and testing for about 45 minutes, the problem was that I did not restart my computer....

#### 6. Searching for fluids simulator c++ tutorials

- **Findings** Found two videos which might help me. But How DO Fluid Simulations Work? and Coding Adventure: Simulating Fluids
- **Findings** First thing in the second video is how to draw a circle. As I am programming in c++ first have to search how to actually show something on my screen.
- **note** Have not yet watched videos.

#### 7. Searching for c++ graphics libraries

- **Findings** Found multiple options for graphics. C# plus c++. SFML. Unity.
- **Activities** Tried to setup c#, unity and SFML.
- **Findings** C# kinda works. It was very annoying to setup on Linux. Have to learn too many things about c# to use it good.
- **Findings** Unity does not work. It makes visualization very easy, but integration with c++ is very hard.
- **Findings** SFML is a good option. It is a c++ library and is easy to use.

## 8. Bouncing ball

- **Note** Decided to first make a bouncing ball in SFML to get a feel for the library.
- **Activity** made a bouncing ball in SFML.

## 9. Grid of bouncing balls

- **Note** Wanted to expand the previous code with multiple bouncing balls.
- **Activity** made a grid of bouncing balls in SFML.

## 10. Clean up code

- **Activity** Cleaned up the code of the grid of bouncing balls.
- **Activity** Added a class for rendering.
- **Activity** Added a class for the physics.
- **Activity** Added submodules for SFML and ImuGui.
- **Activity** Made code more modular.

# Hours Spent

| #                  | Activity                                     | Hours |
|--------------------|----------------------------------------------|-------|
| 1                  | Read brightspace and init project            | 1     |
| 2                  | Google what is OpenCL                        | 0.5   |
| 3                  | Google openCL tutorial                       | 1     |
| 4                  | Trying to use c++                            | 1.0   |
| 5                  | Search and follow basic tutorial of openCL   | 1.0   |
| 6                  | Searching for fluids simulator c++ tutorials | 0.5   |
| 7                  | Searching for c++ graphics libraries         | 6     |
| 8                  | Bouncing ball                                | 1.5   |
| 9                  | Grid of bouncing balls                       | 1.5   |
| 10                 | Clean up code                                | 3.5   |
| 11                 | went from falling balls to equalizer         | 5     |
| <b>Total Hours</b> |                                              | 22.5  |

# Results

- Learned the basics of OpenCL.
- Understood how to use OpenCL for parallel computing.